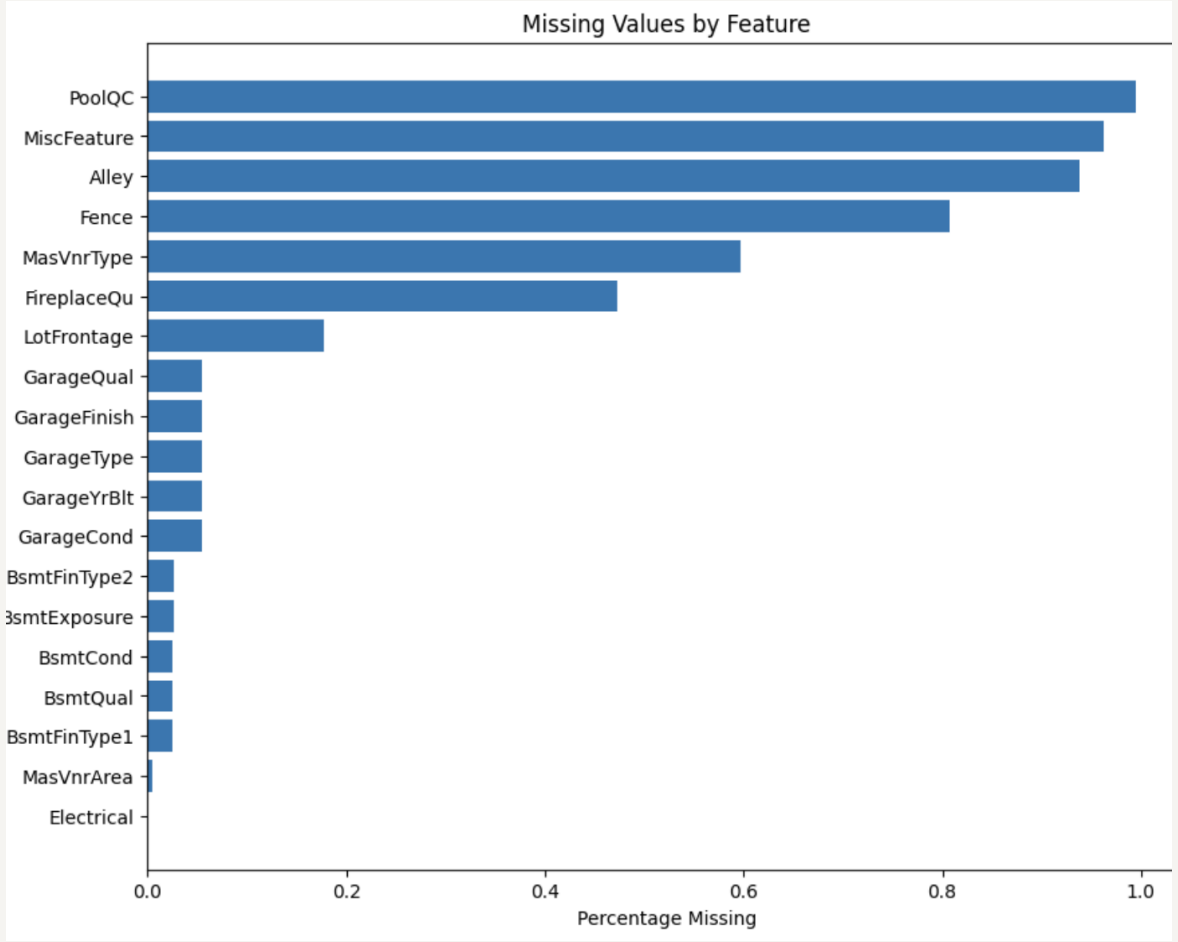
An ML Exploration: Predicting Property Values with Stacking Ensemble Models

Business Context

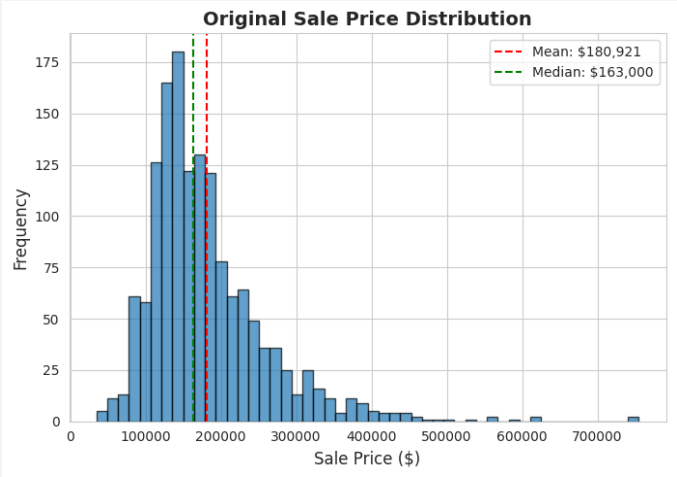
- **Challenge**: Real estate valuation traditionally relies on manual appraisals, which is slow, inconsistent, and difficult to maintain and standardize.
- **Solution**: Automated ML valuation system predicting prices with high accuracy
- Accurate real estate valuation is essential for:
  - Price new listings
  - Reducing days on market
  - Preventing underpricing revenue losses
  - Improving investment decisions

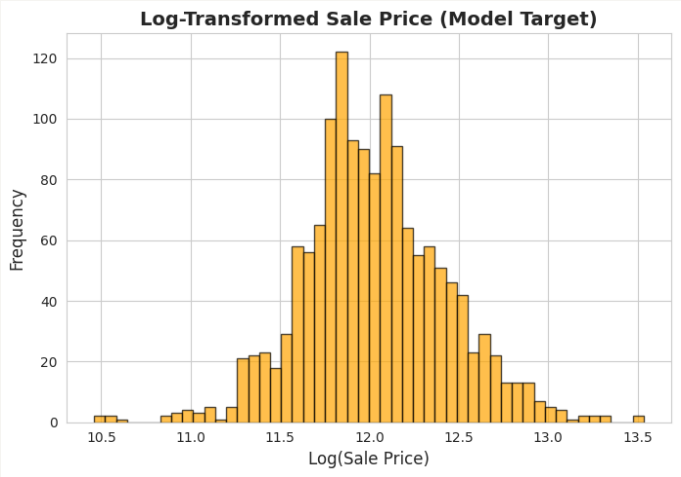
Data Foundation



- Source: The Ames Housing dataset about residential properties in Ames, Iowa
- Scope: 1,460 training properties + 1,459 test properties (81 features)
- Missing Values: Reflects the absence of a particular property feature, not random missingness
- Data Cleaning: Reduced from 1,460 to 1,095 complete training observations (25% reduction)
  - Dropped high-missingness columns (>40% missing)
  - Removed rows with critical missing values

Preprocessing and Feature Engineering





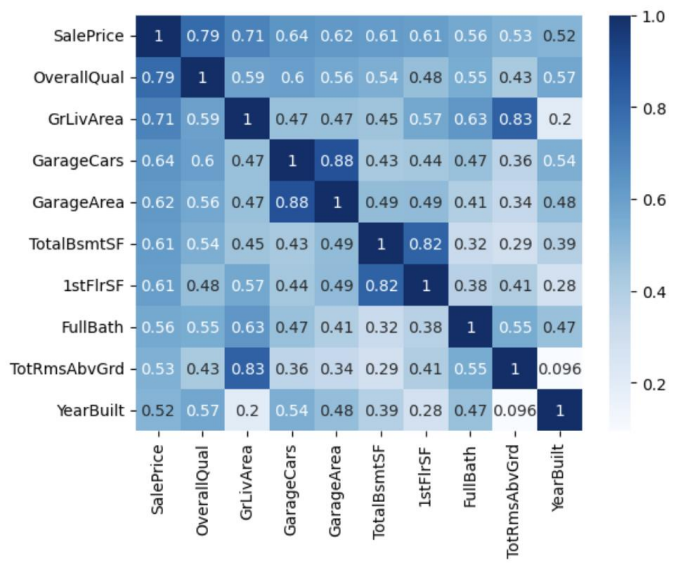
Target Transformation:

- Log transformation (np.log1p) to normalize distribution
- Reduces heteroskedasticity and improves model fit

Encoding: One-hot encoding

Feature Engineering: Created 5 new features to capture housing structure and quality

Outlier Treatment: Capped at 99.5th percentile: SalePrice, TotalBsmtSF, GarageArea



Model Architecture: Stacking Ensemble

Model Development Process:

• Baseline: Dummy Regressor

• Linear Models: Ridge, Lasso, ElasticNet (strong baselines)

• Tree-Based: Random Forest, Gradient Boosting, XGBoost, LightGBM

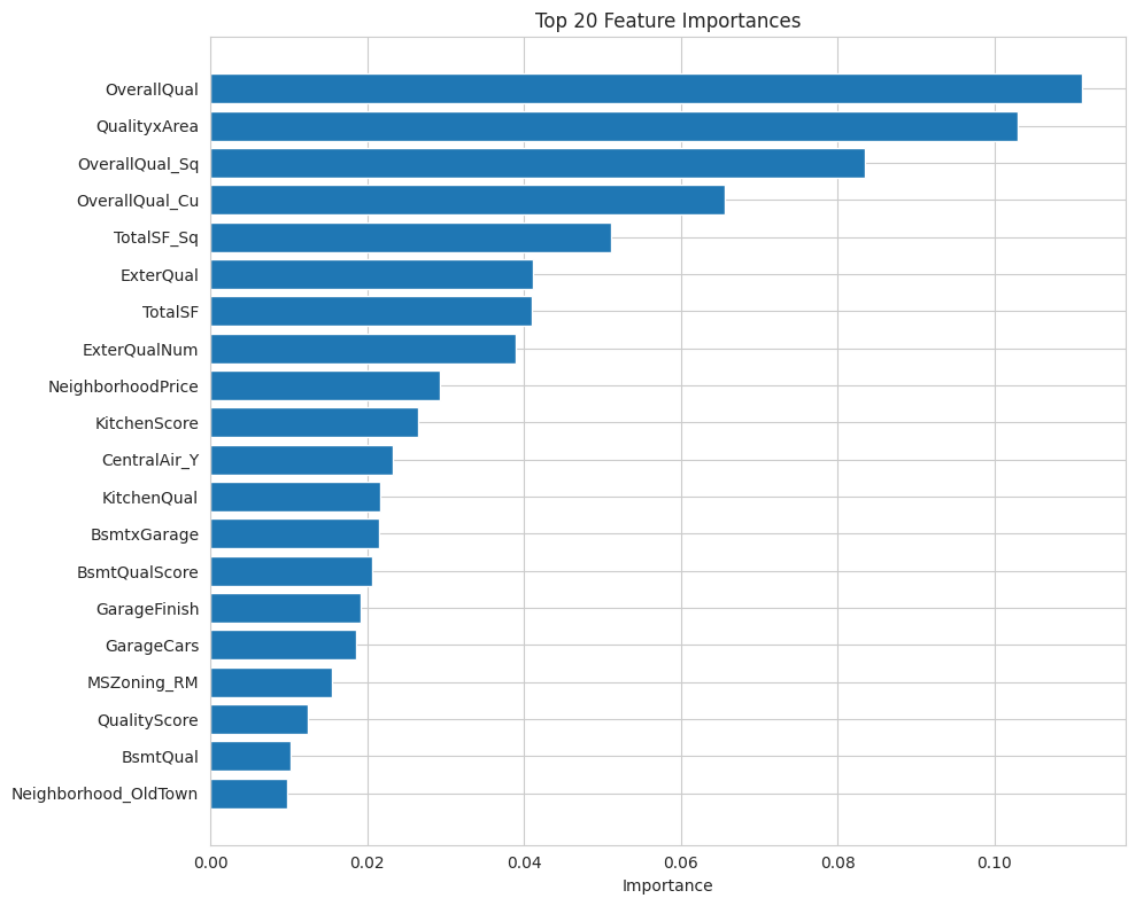
Final Model: Stacking Regressor

- Base Layer (Layer 1):
  - Ridge Regression (alpha=15) - Linear baseline with L2 regularization
  - XGBoost Regressor - Captures non-linear interactions
    - 3,515 estimators, max_depth=4, learning_rate=0.00875
    - Tuned via 5-fold GridSearchCV
- Meta Layer (Layer 2):
  - Linear Regression - Learns optimal weighting of base predictions

- Why Stacking?
  - Combines linear (Ridge) and non-linear (XGBoost) strengths
  - Ridge handles multicollinearity, XGBoost captures feature interactions
  - Meta-learner optimally blends predictions for superior performance

Key Drivers and Business Relevance

- **Quality Over Quantity**: Higher overall material and finish quality has a stronger impact on value
- **Space Optimization**: Focus on above-grade living area for higher market value
- **Garages**: Additional garage capacity homes are consistently valued higher
- **Kitchen Quality**: Higher-rated kitchen quality is strongly associated with higher home values.
- **Key Insight**: Renovations that improve materials, finish, living area, or garage capacity yield the highest ROI.



Key Modeling Insights & Takeaways

Model Key Findings:

- Stacking ensemble outperforms linear models
- XGBoost captures non-linear patterns and interactions
- Ridge provides stable linear baseline and regularization
- Engineered features (TotalSF, Total_Bathrooms) boost accuracy
- Top 1% Kaggle ranking validates production-readiness
- Home quality and size remain the strongest value drivers

Limitations:

- Geographic scope: Ames, Iowa only - requires retraining for other markets
- Temporal: 2006-2010 data may not reflect current market
- Does not account for market cycles, interest rates, or economic conditions
- Reduced training set (1,095 vs 1,460) limits edge case coverage
- Fewer examples for high-end luxury properties

Business Impact

Direct Cost Savings: Automated valuations replace many manual appraisals, saving $400 per property

Reduced Mispricing Losses: 10% accuracy improvement on loan valuations

Operational Efficiency: Appraisal processing time drops 80%, enabling faster underwriting and listing decisions

Total Estimated Annual Value: Combined benefits deliver $1.2M in yearly business impact